Tensor Network Optimisation for Quantum Encoding (Abstract)

Josh Green*[1] and Jingbo Wang[1]

[1]The University of Western Australia

April 4 2025

Abstract

This abstract is a condensed summary of a paper in preparation [1]. We introduce a tensor network optimisation approach to the problem of quantum state preparation. We demonstrate that the Tensor Train (TT) format of tensor networks is an effective dimensionality reduction technique for two-dimensional data, including natural images. We construct an efficient mapping from data approximated by the TT format into optimised quantum circuits for further quantum processing. We showcase the efficacy of this approach by preparing ChestMNIST image data with accuracy exceeding 99.0%.

1 Introduction

Quantum state preparation is the problem of designing algorithms to encode data into the amplitudes of a quantum system. The exponential dimension of the quantum Hilbert space is a critical challenge for the design of efficient state preparation algorithms. Specifically, an n-qubit quantum computer has $\mathcal{O}(2^n)$ degrees of freedom, requiring an exponential-depth circuit to prepare an arbitrary quantum state. We adopt a machine learning approach to quantum state preparation based on efficient tensor networks. We utilise the Matrix Product State (MPS) format, also known as the Tensor Train (TT) format in machine learning communities. The TT format serves as an effective dimensionality reduction technique for classical data. We introduce a tensor train optimisation approach to optimise quantum circuits that prepare approximations of classical data, such as natural images.

2 Methods

2.1 Quantum State Preparation

Let $\mathbf{a} = (a_0, \ldots, a_{N-1})^T \in \mathbb{C}^N$ be a vector of $N = 2^n$ complex numbers, where n denotes the number of qubits in the system. State preparation aims to encode this vector into the amplitudes of a quantum state wavefunction. The state $|\psi\rangle$, known as the target state, is given by:

$$|\psi\rangle = \frac{1}{\|\mathbf{a}\|_2} \sum_{i=0}^{N-1} a_i \, |i\rangle \ , \tag{1}$$

where $\|\mathbf{a}\|_2$, the Euclidean norm of $\mathbf{a}$, is the normalisation coefficient.

2.2 Tensor Train Networks

The MPS or TT format of tensor networks enables the target state $|\psi\rangle$ to be optimally approximated by a chain of n three-index tensors [2]. While any quantum state $|\psi\rangle$ has 2^n elements, we can approximate it using $\mathcal{O}(n\chi^2)$ elements using a TT with bond dimension χ. When the data set obeys an area law of entanglement entropy, meaning that entanglement scales with the area rather than the system's volume, the TT format allows the data set to be accurately approximated with a low bond dimension. The TT format can be viewed as a dimensionality reduction technique that exploits the entanglement entropy structure of quantum states and classical data. The TT format with open boundary conditions is given as:

$$|\psi_{\text{mps}}\rangle = \sum_{i_1,\ldots,i_n=\{0,1\}} A^{[1]i_1}_{\alpha_1} A^{[2]i_2}_{\alpha_1,\alpha_2} \ldots A^{[n-1]i_{n-1}}_{\alpha_{n-2},\alpha_{n-1}} A^{[n]i_n}_{\alpha_{n-1}} |i_1 i_2 \ldots i_n\rangle \tag{2}$$

where $A^{[k]i_k}_{\alpha_{k-1},\alpha_k}$ is the MPS core of dimension $(\alpha_{k-1}, 2, \alpha_k)$ corresponding to the k^{th} qubit, α_{k-1} is the left virtual index, α_k is the right virtual index, and i_k is the physical index. The left and right virtual indices connect each MPS core to its neighbours, with the left virtual index of $A^{[0]}$ and the right virtual index of $A^{[n]}$ set to 1. The bond dimension of the MPS is defined as $\chi := \max_k \alpha_k$.

2.3 Tensor Train Optimisation

We can express the quantum state preparation problem as a minimisation problem. We aim to minimise the following objective function:

$$L(\theta) = 1 - |\langle \psi_{\text{mps}} | U(\theta) | 0 \rangle|^2 \tag{3}$$

where $|\psi_{\text{mps}}\rangle$ is the target state and $U(\theta)$ is a quantum circuit parameterised by the set of variational parameters θ. We minimise the objective function $L(\theta)$ w.r.t θ. Gradient descent is computed via the quasi-Newton L-BFGS-B method. The parameters at iteration $t+1$ are updated as:

$$\theta^{(t=1)} = \theta^{(t)} - \eta H_t \nabla L(\theta^{(t)}) \tag{4}$$

where θ is the set of parameters, η is the learning rate, and H_t is the approximate Hessian matrix at iteration t. The gradients are computed with automatic differentiation, and we set the learning rate $\eta = 0.05$.

Since the quantum circuit $U(\theta)$ has $\mathcal{O}(2^n)$ degrees of freedom, generic choices of the quantum circuit parameterization lead to the vanishing gradient problem [3]. Instead, we utilise an algorithm to approximately map $|\psi_{\text{mps}}\rangle$ into a quantum circuit. This algorithm, known as the Matrix Product Disentangler (MPD) and introduced by Shi-Ju Ran in 2020 [4], analytically prepares an approximation of $|\psi_{\text{mps}}\rangle$, though it lacks rigorous convergence guarantees. The MPD circuit has a hyperparameter, L, denoting the number of circuit layers. We use the quantum circuit generated by the MPD algorithm to initialise the parameters $\theta^{(0)} = \theta_{\text{MPD}}$. If the initial parameters $\theta^{(0)}$ have non-trivial overlap with the global minimum, the severe vanishing gradient problem is significantly mitigated.

3 Experiments

We apply tensor network optimisation (TNO) with parameters initialised by the MPD algorithm to prepare quantum states representative of 128×128 ChestMNIST images on $n = 14$ qubits. Natural images are widely suspected to follow area laws of entanglement entropy, which helps explain the success of deep learning for image classification [5]. We can approximately prepare the ChestMNIST data with an accuracy exceeding 99.0% on linear-depth quantum circuits, significantly reducing circuit depth when using TNO (Figure 1). This is in contrast to $\mathcal{O}(2^n)$-depth circuits required to *exactly* encode images of dimension $2^{\frac{n}{2}} \times 2^{\frac{n}{2}}$.

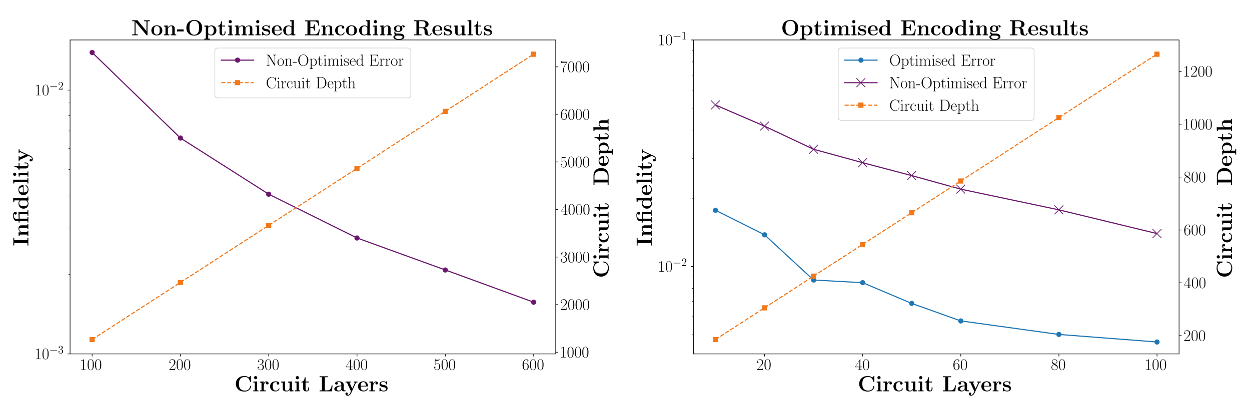

Figure 1: Image encoding results using the (non-optimised) MPD and (optimised) MPD+TNO algorithms.

References

[1] J. Green and J. B. Wang, "Quantum encoding of structured data with matrix product states," 2025. [Online]. Available: https://arxiv.org/abs/2502.16464

[2] R. Orús, "A practical introduction to tensor networks: Matrix product states and projected entangled pair states," *Annals of Physics*, vol. 349, pp. 117–158, 2014. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0003491614001596

[3] M. Cerezo, A. Sone, T. Volkoff, L. Cincio, and P. J. Coles, "Cost function dependent barren plateaus in shallow parametrized quantum circuits," *Nature Communications*, vol. 12, p. 1791, Mar. 2021. [Online]. Available: https://doi.org/10.1038/s41467-021-21728-w

[4] S.-J. Ran, "Encoding of matrix product states into quantum circuits of one- and two-qubit gates," *Phys. Rev. A*, vol. 101, p. 032310, Mar 2020. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevA.101.032310

[5] S. Lu, M. Kanász-Nagy, I. Kukuljan, and J. I. Cirac, "Tensor networks and efficient descriptions of classical data," *Phys. Rev. A*, vol. 111, p. 032409, Mar 2025. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevA.111.032409